Youngmin Kim

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Professional Summary

*A Ph.D. Student in Artificial Intelligence with a research focus on **video understanding** and **human-interacted AI**. My work investigates how AI systems can interpret complex visual information from videos and connect this understanding to human perception and communication, with particular interest in **nonverbal expressions and sign language**.*

Education

Yonsei University — Seoul, South Korea
Ph.D. Student in Artificial Intelligence — Sep. 2023 – Present
Ph.D. Advisor: Prof. Seong Jae Hwang
M.S. Advisor: Prof. Youngjae Yu

Incheon National University — Incheon, South Korea
B.S. in Economics & Computer Science — Mar. 2016 – Mar. 2023
C.S. Advisor: Prof. Hyeongboo Baek
Economics Advisor: Prof. Sangyoen Hwang

Publications (*: Equal Contribution)

Conference Publications

1. **QWERTY: Training-Free Motion Control via Query-Warped Video Diffusion Transformers** [paper]
   Kyobin Choo, **Youngmin Kim**, Hyunkyung Han, Geunrip Park, Chanyoung Kim, Sunyoung Jung, Seong Jae Hwang
   European Conference on Computer Vision (**ECCV**), 2026
2. **ViKey: Enhancing Temporal Understanding in Videos via Visual Prompting** [paper]
   Yeonkyung Lee*, Dayun Ju*, **Youngmin Kim**, Seil Kang, Seong Jae Hwang
   IEEE / CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2026
3. **MAVL: A Multilingual Audio-Video Lyrics Dataset for Animated Song Translation** [paper]
   Woohyun Cho, **Youngmin Kim**, Sunghyun Lee, Youngjae Yu
   Empirical Methods in Natural Language Processing (**EMNLP**), 2025
4. **Speaking Beyond Language: A Large-Scale Multimodal Dataset for Learning Nonverbal Cues from Video-Grounded Dialogues** [paper]
   Youngmin Kim*, Jiwan Chung*, Jisoo Kim, Sunghyun Lee, Sangkyu Lee, Junheok Kim, Cheoljong Yang, Youngjae Yu
   Association for Computational Linguistics (**ACL**), 2025
5. **Scalp Diagnostic System With Label-Free Segmentation and Training-Free Image Translation** [paper]
   Youngmin Kim*, Saejin Kim*, Hoyeon Moon, Youngjae Yu, Junhyug Noh
   Medical Image Computing and Computer Assisted Intervention (**MICCAI**), 2025
6. **A 2-Stage Model for Vehicle Class and Orientation Detection with Photo-Realistic Image Generation** [paper]
   Youngmin Kim*, Donghwa Kang*, Hyeongboo Baek
   IEEE International Conference on Big Data (**IEEE BigData**), 2022
7. **Pedestrian Accident Prevention Model Using Deep Learning and Optical Flow** [paper]
   Youngmin Kim, Gyujin Jang, Hyeonjae Bae, Youngnam Kim, Jinpyung Kim
   Korea Computer Congress (**KCC**), 2021
   Best Paper Award (Undergraduate Session)

Journal Publications

1. **AI-Powered Real-Time System for Automated Concrete Slump Prediction via Video Analysis** [paper]

Youngmin Kim, Giyeong Oh, Kwangsoo Youm*, Youngjae Yu*
Automation in Construction (**AiC**), 2026, IF: 11.5

2. **Preprocessing for Keypoint based Sign Language Translation without Glosses** [paper]
Youngmin Kim, Hyeongboo Baek.
Sensors, 2023, IF: 3.8

3. **A Study of Tram-Pedestrian Collision Prediction Method Using YOLOv5 and Motion Vector** [paper]
Youngmin Kim, Hyeonuk Ahn, Heegyun Jeon, Jinpyung Kim, Gyujin Jang, Hyeonchyeol Hwang
Korea Information Processing Society (KIPS), 2021

4. **Optical Flow Estimation Techniques and Recent Research Trends** [paper]
Youngmin Kim, Hyeonuk Ahn, Jinpyung Kim
Korea Information Processing Society Reviews, 2021

Pre-prints

1. **BLANC: Adaptive Temporal Scheduling for Sign Language Production** [paper]
Youngmin Kim, Seoyeong Hwang, Sujung Hong, Seong Jae Hwang
Pre-print

2. **Towards Continuous Sign Language Conversation from Isolated Signs** [paper]
Youngmin Kim, Kyobin Choo, Jiwoo Park, Minseo Kim, Chanyoung Kim, Junhyeok Kim, Seong Jae Hwang
Pre-print

3. **ViT Attention Weight as a Hierarchical Tree** [paper]
Junhyeok Kim, Minseo Kim, Seil Kang, **Youngmin Kim**, Seong Jae Hwang
Pre-print

4. **Sparse Feature Analysis and Control for Robust Vision-Language-Action Models** [paper]
Minseo Kim, Junhyeok Kim, Jinyeong Kim, **Youngmin Kim**, Seong Jae Hwang
Pre-print

5. **Starve the MLP, Feed the Attention: Resolving Vision-Knowledge Conflict in VideoLLMs** [paper]
Chanyong Yoon, **Youngmin Kim**, Seong Jae Hwang
Pre-print

6. **CogViD: A Comprehensive Cognitive Benchmark for Evaluating Video MLLMs** [paper]
Khawar Islam, Muhammad Zaigham Zaheer, **Youngmin Kim**, Soyeong Jeong, Basim Azam, Muzammal Naseer, Naveed Akhtar
Pre-print

Experience

MIR Lab, Yonsei University Mar. 2022 – Mar. 2023
Undergraduate Researcher

RAISE Lab, Incheon National University Dec. 2021 – Feb. 2023
Undergraduate Researcher

CVAI Lab, Advanced Institute of Convergence Technology (AICT) Sep. 2020 – Aug. 2021
Research Assistant

Scholarly Activities

Invited Speaker

- **Samsung AI Education** 2023
Introduction to Multimodal Research

Teaching Assistant

- Introduction to AI Research (Yonsei University) 2024

- Reinforcement Learning (Yonsei University) 2024
- Basic of PyTorch (Upstage) 2023

Academic Service

Conference Reviewer

- **ICLR**

Journal Reviewer

- **IEEE Transactions on Image Processing**

Selected Projects

LIG Defense&Aerospace Aug. 2026 – Present

Synthetic IIR Dataset Generation

Hemily Hospital Jun. 2026 – Present

Medical Foundation Model (Dr.Seong Joo Hwang LLM)

LG Electronic Mar. 2025 – Aug. 2025

Multimodal Contents Recommendation System

GS E&C Dec. 2023 – Feb. 2025

Video-based Slump Prediction

NCSOFT Jan. 2024 – Jun. 2024

Multimodal Conversational Agent

Awards & Scholarship

Doctoral Excellence Scholarship (Science and Engineering) 2026
Selected recipient, Korea Student Aid Foundation (KOSAF)

AI-Hub Idea Challenge Competition 2021
1st prize, South Korea's National Information Society Agency (NIA)

KCC 2021 Undergraduate Paper Competition 2021
1st prize, Korean Institute of Information Scientists and Engineers (KIISE)

KED 2021 Industrial Innovation Big Data Platform Competition 2021
2nd prize, Korea Enterprise Data (KED)

Patents

Method and apparatus for avoiding collision between vehicle and object, computer program 2023
KR102533419B1

Electronic apparatus and method for searching distress, unmanned aerial vehicle, computer program 2023
KR102587311B1

Apparatus and method for analyzing data, apparatus and method for predicting abnormality, computer program 2023
KR102480394B1